

Week 2

This week we will:

2.1: Learn about compound interest

2.1 a&b: Learn how to use a financial calculator to calculate FV and PV

2.2: Learn about Effective Annual Rate

2.3: Learn about Net Present Value and practise to calculate the "added value" from an investment to shareholders' wealth

Learning Outcomes

At the end of this module you will:

SLO1: describe the principles and objectives of financial decision-making

SLO2: apply time-value of money techniques to the valuation of financial instruments and financial securities

SLO4: evaluate a corporate investment proposal and provide a recommendation to accept or reject it

Compound Interest

- Compound Interest has interest earned added to the principal to form a new principal on which the next calculation of interest is based
 - interest on interest or compounding
 - lots of simple interest calculations
- The compounding period can be any length of time
 - Yearly
 - half-yearly
 - Quarterly
 - Monthly
 - Daily

Future Value

- Alice deposits \$1,000 today in a savings account that pays interest once a year.
- How much will Alice have in three years' time if the interest rate is 12% pa?



$$FV = PV (1 + i)^n$$

- i = interest rate per period (12% pa)
- n = number of compounding periods (3 years)
- PV = the original principal (\$1,000)

Present Value

- We can rearrange the formula to calculate present value

$$PV = \frac{FV}{(1+i)^n}$$

Example

- Suppose a stock currently pays a dividend of \$1.10, which is expected to grow at 40% per year for the next five years.
- What will the dividend be in five years?

$$FV = C_0 \times (1 + r)^T$$

$$\$5.92 = \$1.10 \times (1.40)^5$$

Future Value and Compounding

- Notice that the dividend in year five, \$5.92, is considerably higher than the sum of the original dividend plus five increases of 40-percent on the original \$1.10 dividend:

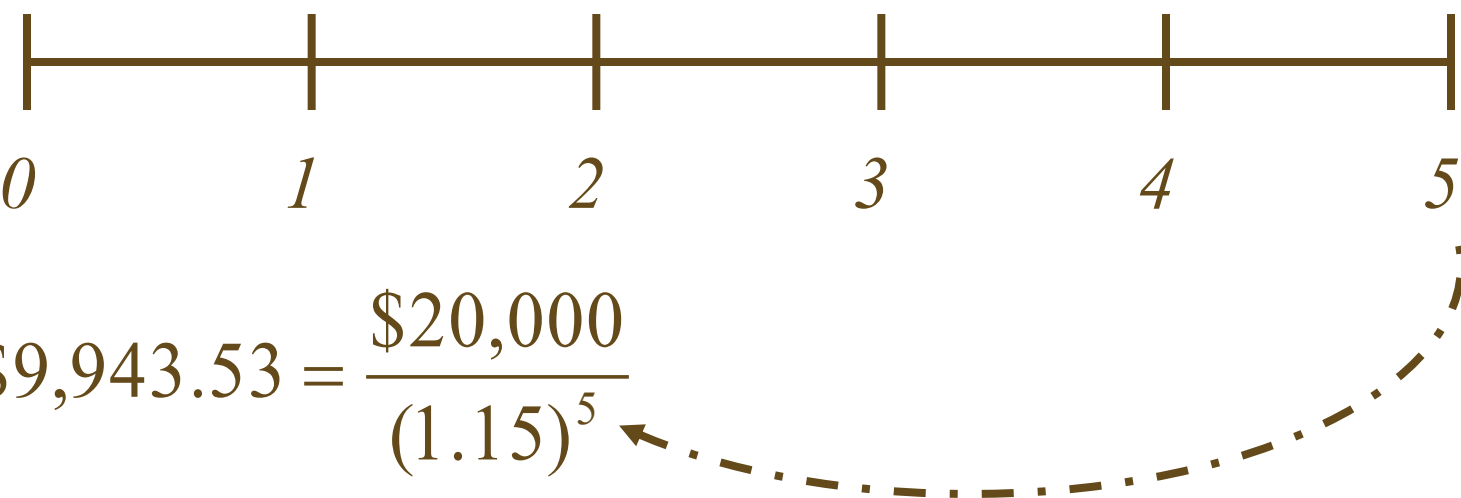
$$\$5.92 > \$1.10 + 5 \times [\$1.10 \times .40] = \$3.30$$

This is due to **compounding**.

Example: Multiperiod Present Value and Discounting

- How much would an investor have to set aside today in order to have \$20,000 five years from now if the current rate is 15%?

PV \$20,000


$$\$9,943.53 = \frac{\$20,000}{(1.15)^5}$$

Solve for Any Variable

- Examples thus far have offered the time and interest rate and solved for PV or FV
- Keep in mind that there are four variables:
 - PV
 - FV
 - T
 - R
- If you have any three you can solve for the fourth
- The math can become cumbersome
 - Financial Calculators and Spreadsheets are very helpful

Example: Solving for Time

If we deposit \$5,000 today in an account paying 10%, how long does it take to grow to \$10,000?

$$FV = C_0 \times (1 + r)^T \qquad \$10,000 = \$5,000 \times (1.10)^T$$

$$(1.10)^T = \frac{\$10,000}{\$5,000} = 2$$

$$\ln(1.10)^T = \ln(2)$$

$$T = \frac{\ln(2)}{\ln(1.10)} = \frac{0.6931}{0.0953} = 7.27 \text{ years}$$

Example: Solving for Required Return

Assume the total cost of a college education will be \$50,000 when your child enters college in 12 years. You have \$5,000 to invest today. What rate of interest must you earn on your investment to cover the cost of your child's education?

About 21.15%.

$$FV = C_0 \times (1 + r)^T \qquad \$50,000 = \$5,000 \times (1 + r)^{12}$$

$$(1 + r)^{12} = \frac{\$50,000}{\$5,000} = 10 \qquad (1 + r) = 10^{1/12}$$

$$r = 10^{1/12} - 1 = 1.2115 - 1 = .2115$$

Frequency of Compounding

- Interest rates are normally quoted as per annum
- But the compounding frequency is not always annual
- A nominal rate (a state annual rate) is when interest is compounded more frequently than annually
- If the compounding period is not annual the rate used in the calculations must be adjusted to the rate per compounding period
 - E.g., 16% pa compounded monthly
 - This nominal rate is not the same as 16% pa

Example

- ❑ What is the compounding rate for each time period for an 18% annual interest rate, compounding monthly?
- ❑ The number of compounding periods each year is 12
- ❑ Rate per period = $18\% \div 12 = 1.5\%$

Simple Interest vs Compound Interest

Simple interest	Compound interest
$FV = PV \times (1 + i \times n)$	$FV = PV \times (1 + i)^n$
$PV = \frac{FV}{(1 + i \times n)}$	$PV = \frac{FV}{(1 + i)^n}$

- When to use Compound interest ?
 - Any calculation with a time length of 12 months
- When to use Simple interest ?
 - Any calculation with a time length of 12 months

Effective Annual Rate

- To **compare rates with different compounding periods** we need to convert both rates to an equivalent annual rate
- This is the **effective annual rate (EAR)**

$$\text{EAR} = (1+i)^m - 1$$

i = interest rate per period

m = compounding periods per year

Compounding Periods

- For example, if you invest \$50 for 3 years at 12% compounded semi-annually, your investment will grow to

$$FV = \$50 \times \left(1 + \frac{.12}{2}\right)^{2 \times 3} = \$50 \times (1.06)^6 = \$70.93$$

Effective Annual Rates of Interest

A reasonable question to ask in the above example is “what is the effective *annual* rate of interest on that investment?”

$$FV = \$50 \times \left(1 + \frac{.12}{2}\right)^{2 \times 3} = \$50 \times (1.06)^6 = \$70.93$$

The Effective Annual Rate (EAR) of interest is the annual rate that would give us the same end-of-investment wealth after 3 years:

$$\$50 \times (1 + EAR)^3 = \$70.93$$

Effective Annual Rates of Interest

$$FV = \$50 \times (1 + EAR)^3 = \$70.93$$

$$(1 + EAR)^3 = \frac{\$70.93}{\$50}$$

$$EAR = \left(\frac{\$70.93}{\$50} \right)^{1/3} - 1 = .1236$$

So, investing at 12.36% compounded annually is the same as investing at 12% compounded semi-annually.

Effective Annual Rates of Interest

- Find the Effective Annual Rate (EAR) of an 18% APR loan that is compounded monthly.
- What we have is a loan with a monthly interest rate rate of $1\frac{1}{2}\%$.
- This is equivalent to a loan with an annual interest rate of 19.56%.

$$\left(1 + \frac{r}{m}\right)^m = \left(1 + \frac{.18}{12}\right)^{12} = (1.015)^{12} = 1.1956$$

Continuous Compounding

- The general formula for the future value of an investment compounded continuously over many periods can be written as:

$$FV = C_0 \times e^{rT}$$

Where

C_0 is cash flow at date 0,

r is the stated annual interest rate,

T is the number of years, and

e is a constant that is approximately equal to 2.718. e^x is a key on your calculator.

Net Present Value (NPV)

- NPV is one of the indexes that indicates the value added to the firm from an investment. It takes into account the cash outflows and cash inflows of the investment in today's value
- **NPV = Sum of Present Values of all cash flows in today's value**

Example

- Meryl is planning for an overseas trip.
- Meryl already has \$7,000 which she deposits today and will invest a further \$10,000 in six months time
- What is the accumulated value in two years?
- The interest rate is 7.5% pa compounded half-yearly.

Example

- 7,000 10,000
FV?
- $i = 7.5\% \div 2 = 3.75\%$
- $FV_{7000} = 7,000(1+0.0375)^4 = 8,110.55$
- $FV_{10000} = 10,000(1+0.0375)^3 = 11,167.71$
- Meryl has \$19,278.26 in two years

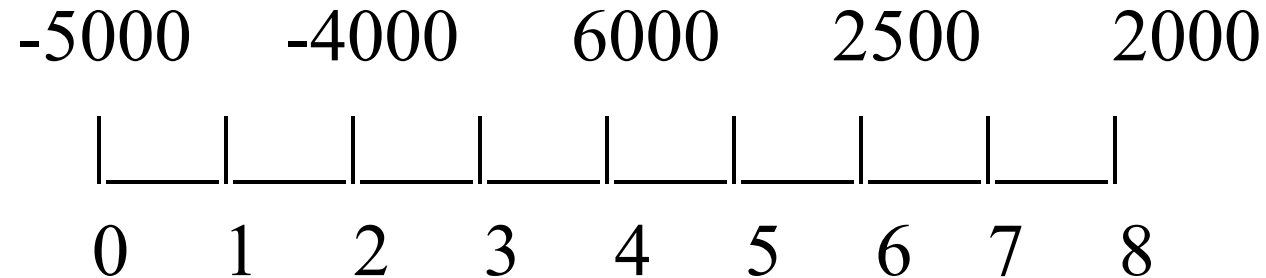
Practice

- Thunderbirds Production Company (TPC) is offering investors an opportunity to invest in its movie production business
- TPC is asking investors to invest \$5,000 now and \$4,000 in two years' time
- TPC has projected the cash inflows from its movie to investors would be \$6,000 in year four, \$2,500 in year six and a final amount in year eight of \$2,000

Practice cont.

- Investments with similar risks have a return of 20% pa.
 - Is it worth investing in the TPC project?

• **CASH FLOWS:**



Practice cont.

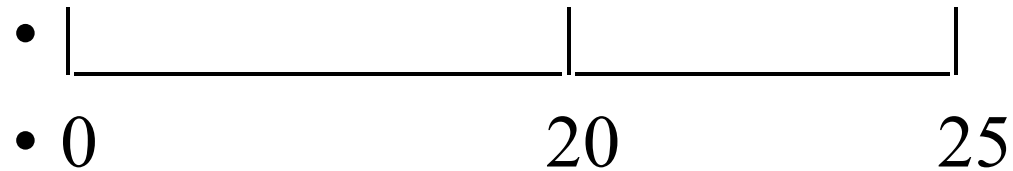
- $-5,000 = -5,000$ PV of Year 0 cash flow
- $-4,000(1.2)^{-2} = -2,778$ PV of Year 2 cash flow
- $+6,000(1.2)^{-4} = 2,894$ PV of Year 4 cash flow
- $+2,500(1.2)^{-6} = 837$ PV of Year 6 cash flow
- $+2,000(1.2)^{-8} = 465$ PV of Year 8 cash flow
- Total of Present Values = $-\$3,582$
- Thunderbirds is a poor investment because the PVs of the positive cash flows are less than the PVs of the negative cash flows!

Practice

- Bert and Beryl want to establish a fund that will pay \$10,000 to each of their two grandchildren, Peter and Mary, when they turn 21
- Mary turns 21 in ten years' time and Peter turns 21 in twelve-and-a-half years' time.
- If the fund's interest rate is 8% pa. compounded half-yearly, what amount should Bert and Beryl deposit today?

Practice cont.

\$10,000 \$10,000



- $i = 8\% / 2 = 4\%$
- FV = \$10,000 in both cases
- PV of Mary's payment = $10,000 / (1 + .04)^{20}$
- PV of Peter's payment = $10,000 / (1 + .04)^{25}$
- $PV = 4,563.87 + 3,751.17$
 $= \$8,315.04$